

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

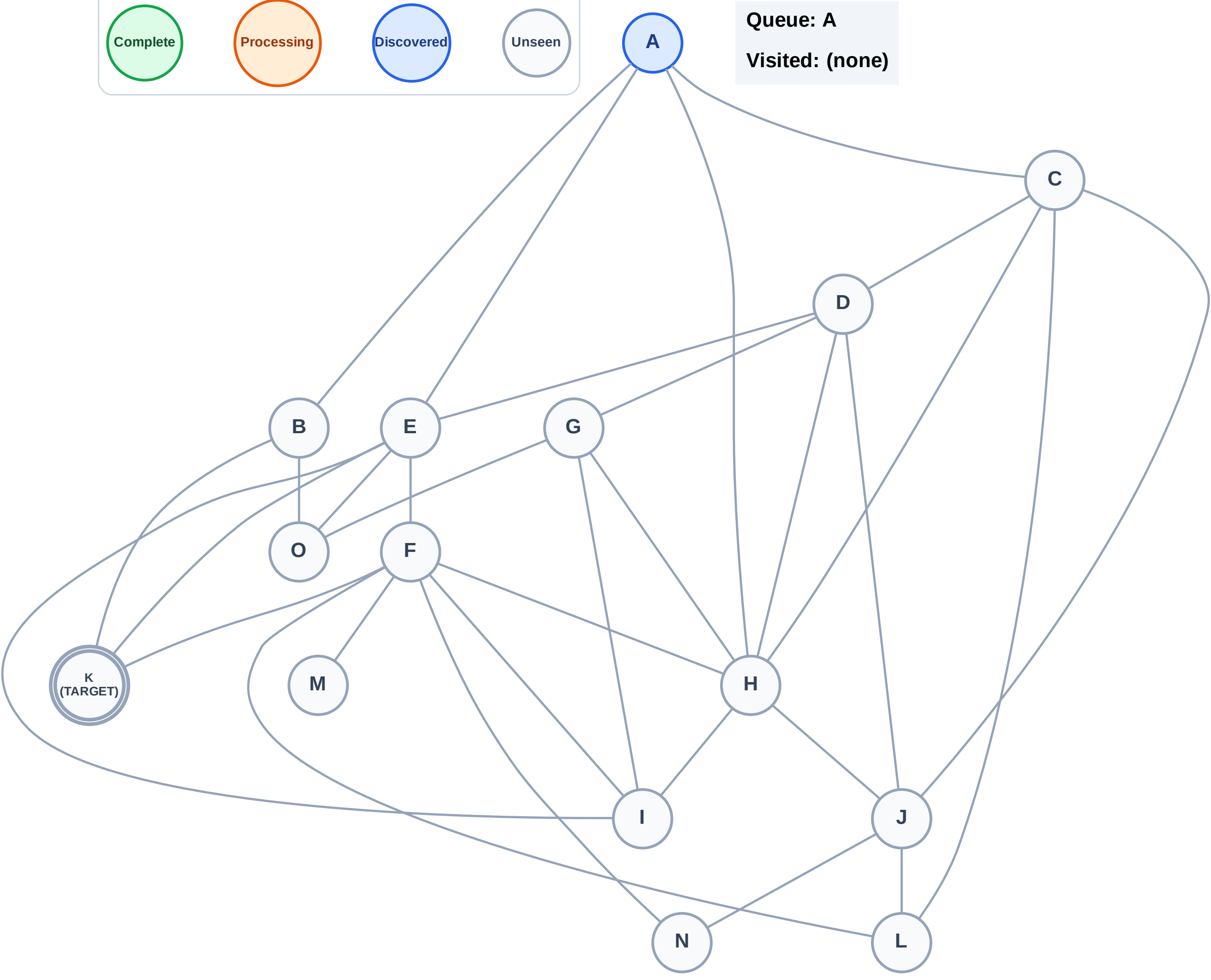
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

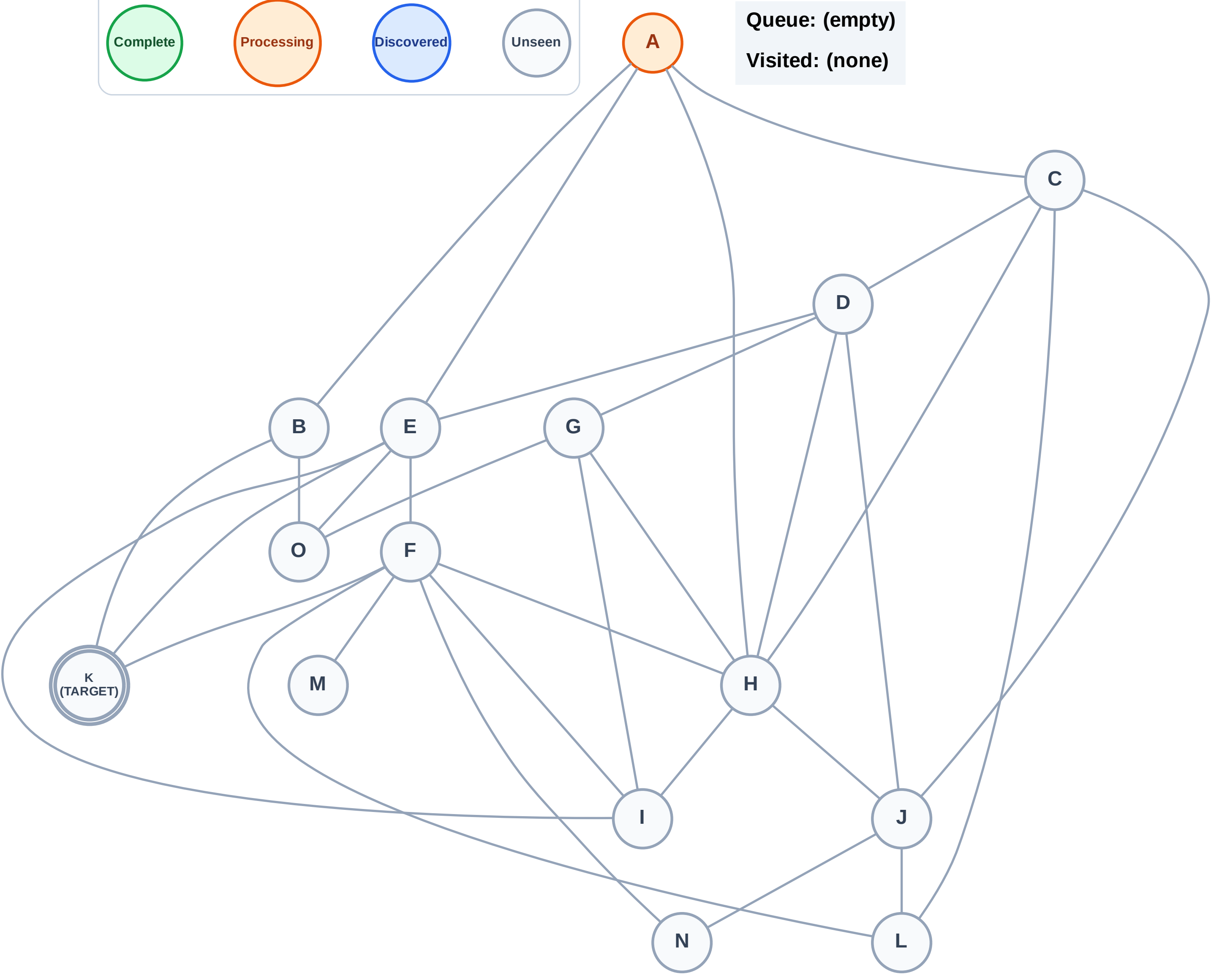
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, H from A

Legend

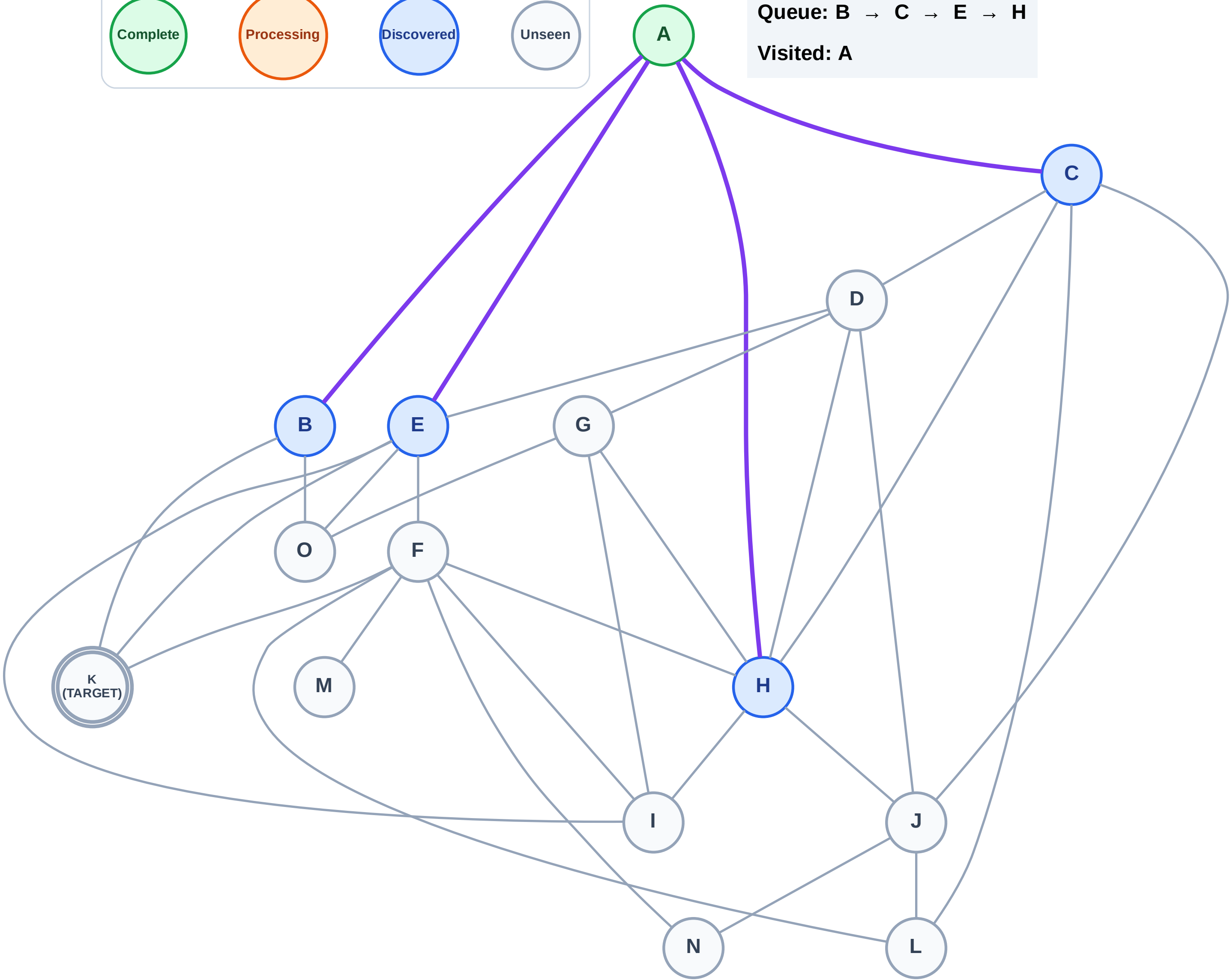
Complete

Processing

Discovered

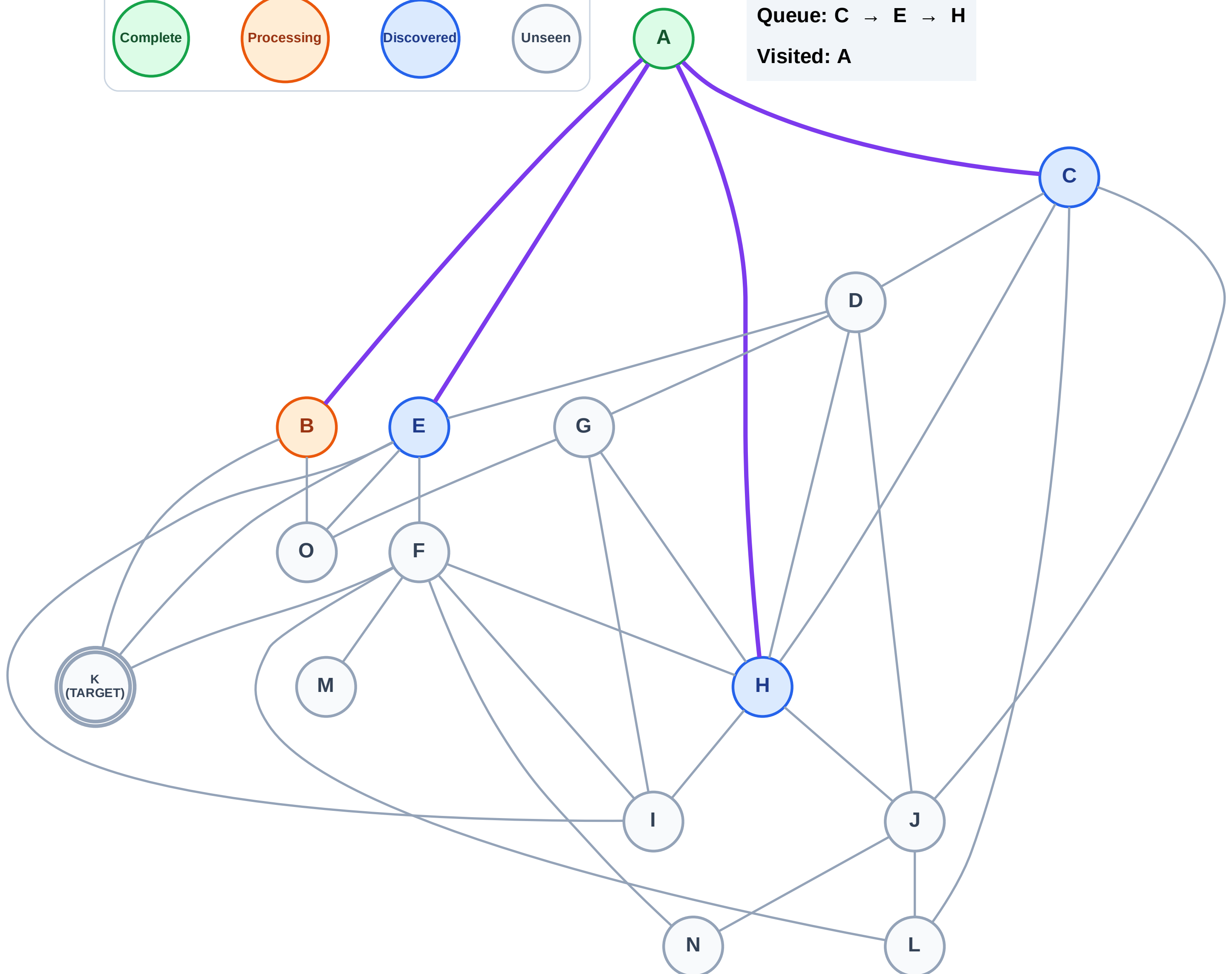
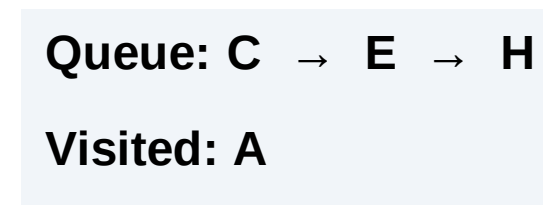
Unseen

Queue: B → C → E → H
Visited: A



Step 4: Process node B

Step 4: Process node B



BFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend

Complete

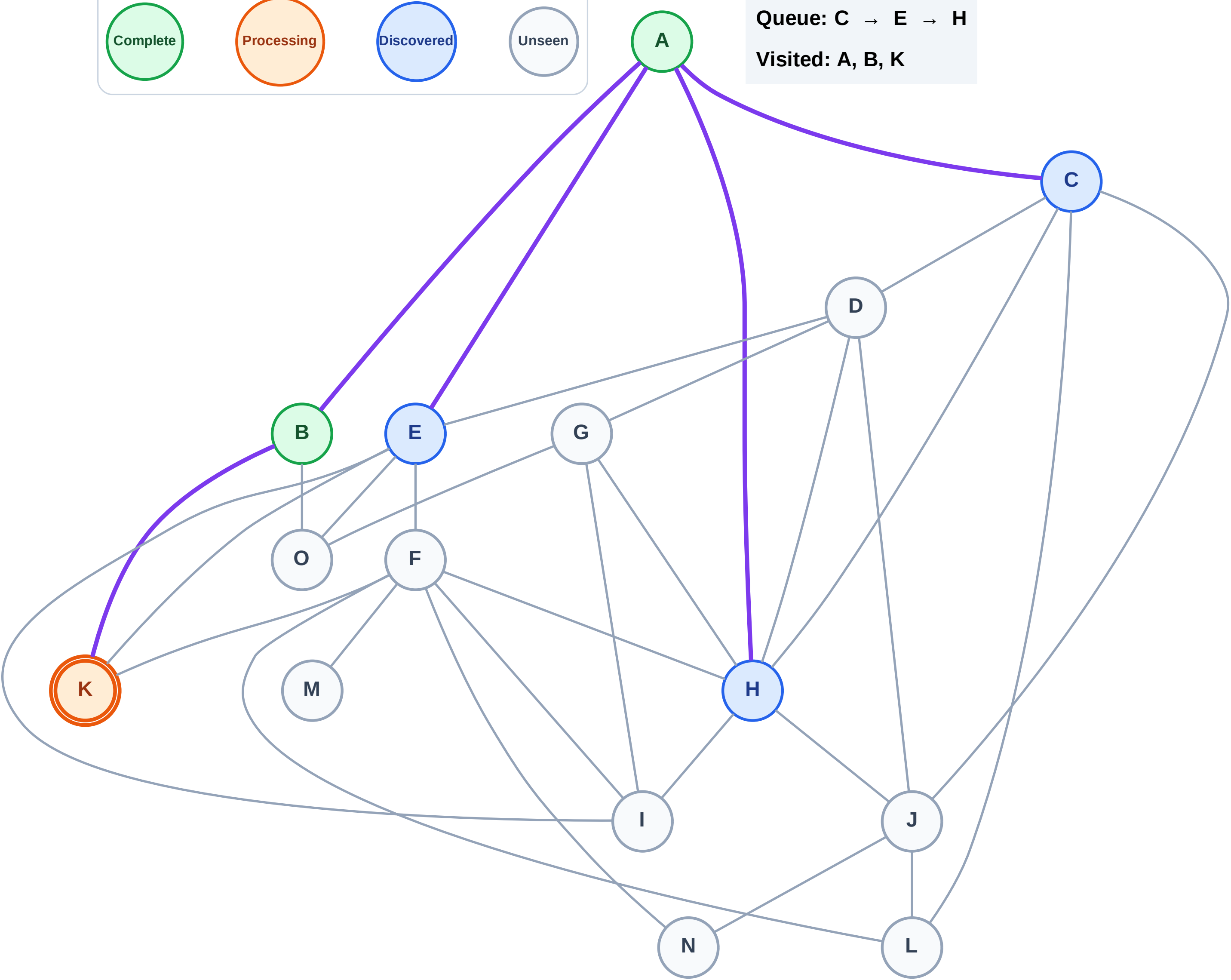
Processing

Discovered

Unseen

Queue: C → E → H

Visited: A, B, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

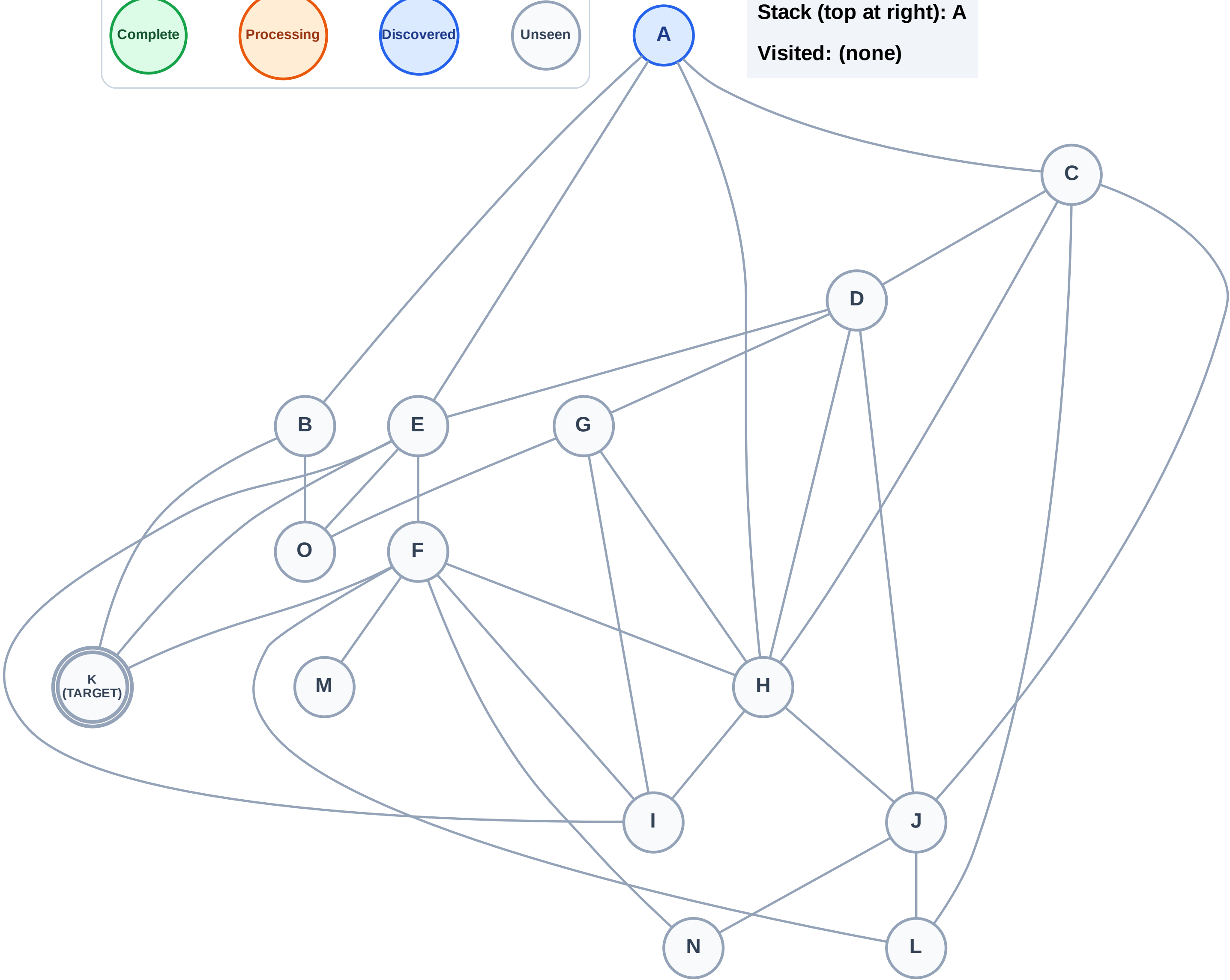
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

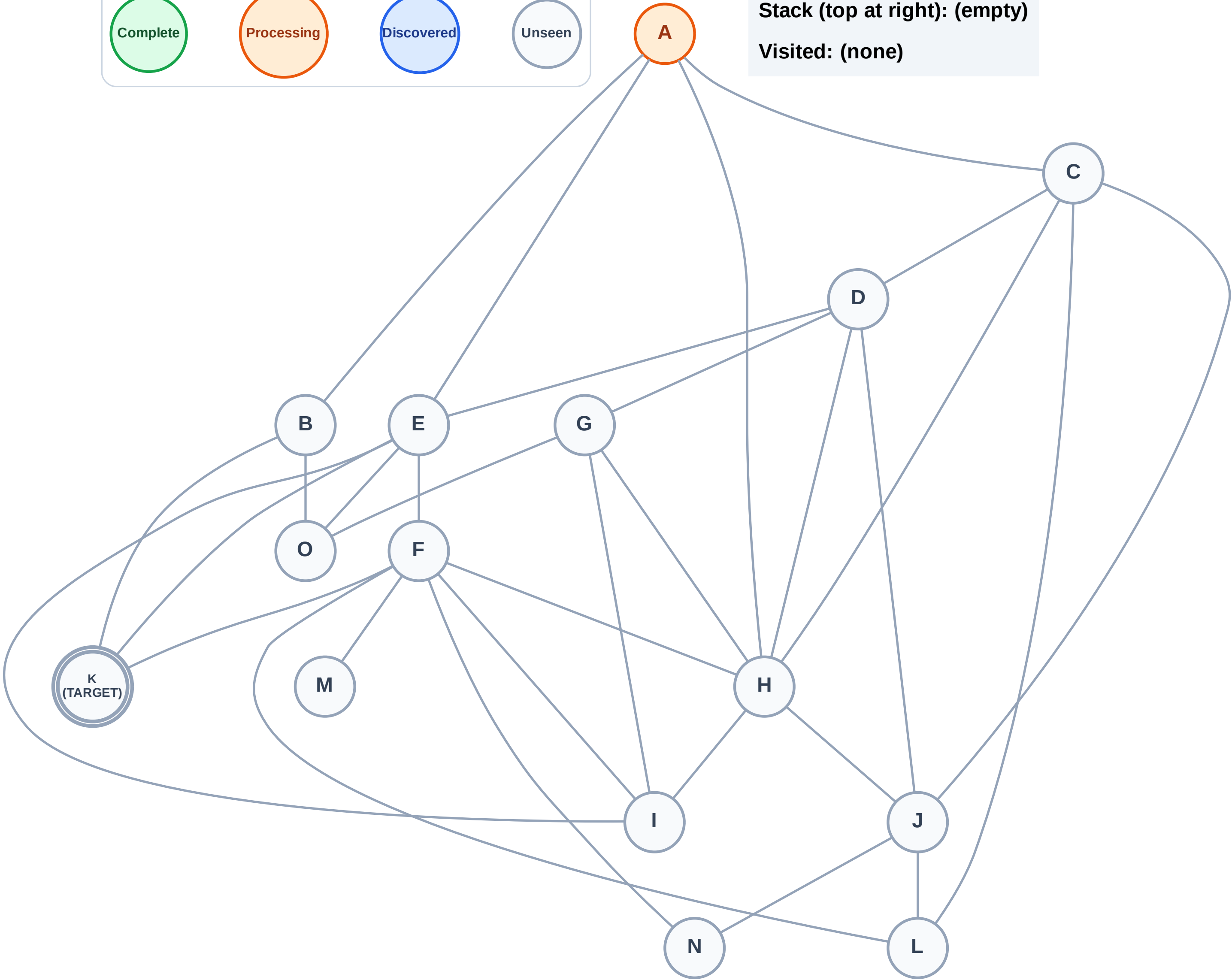
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, E, C, B from A

Legend

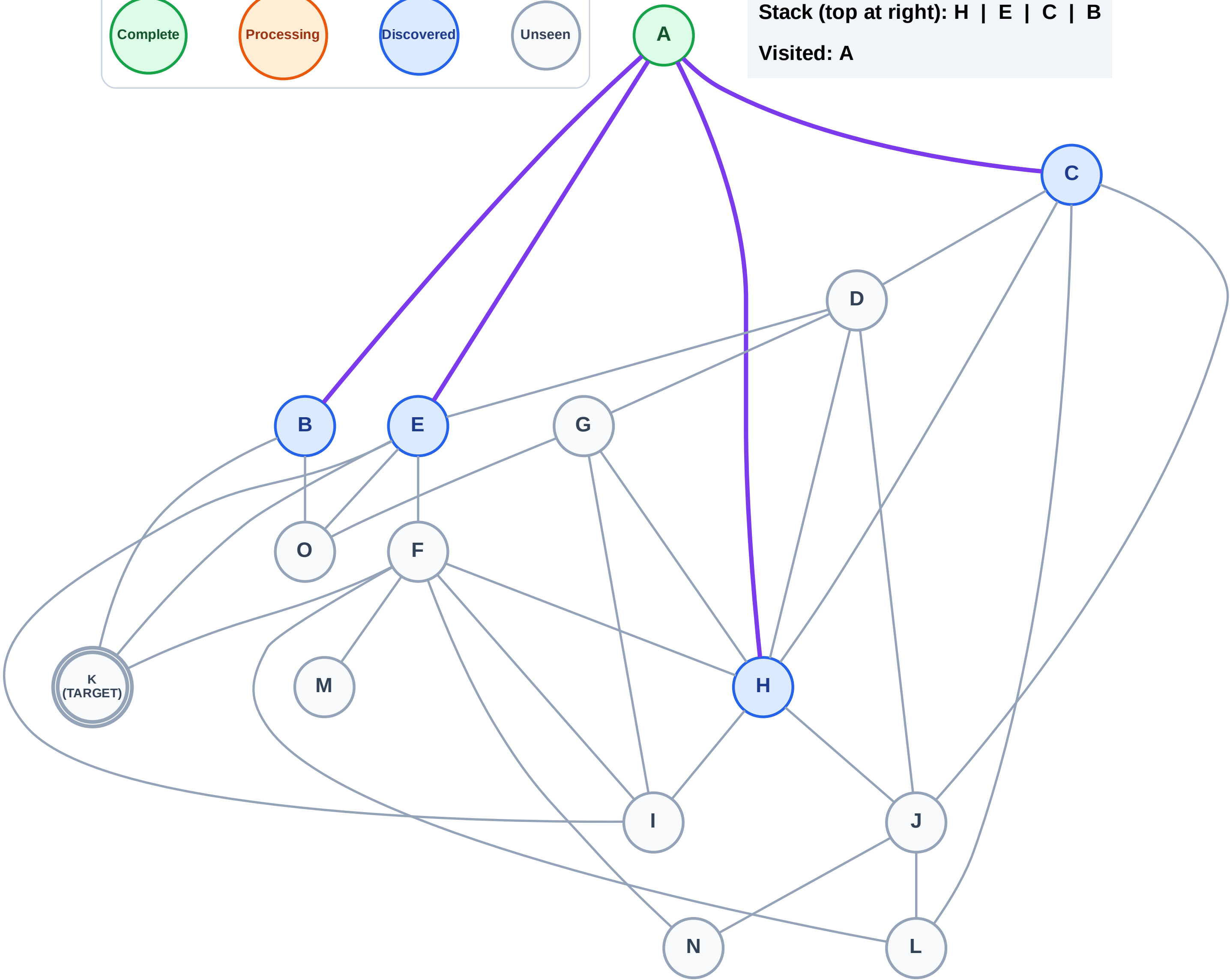
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

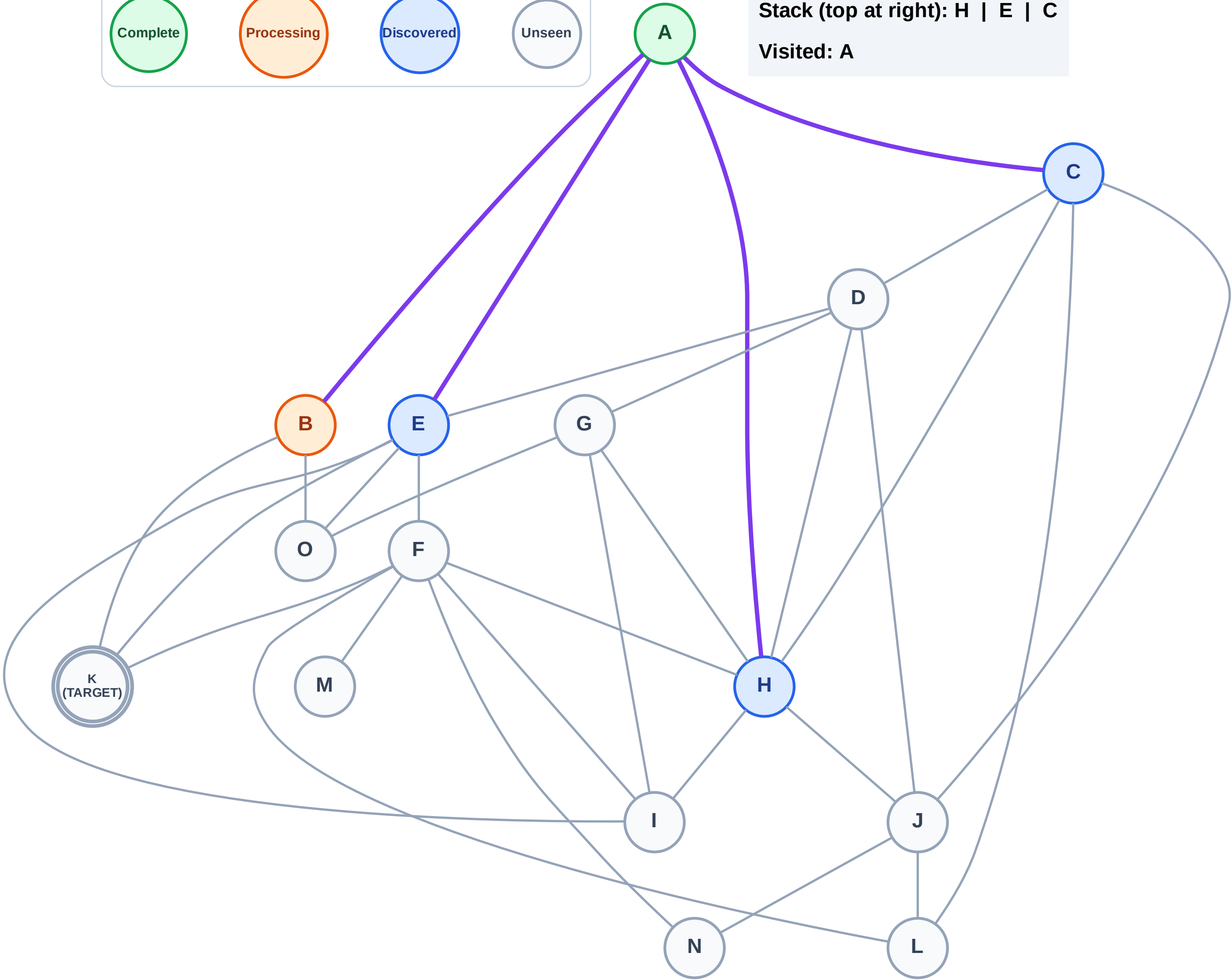
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | C
Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | C | O

Visited: A, B, K

